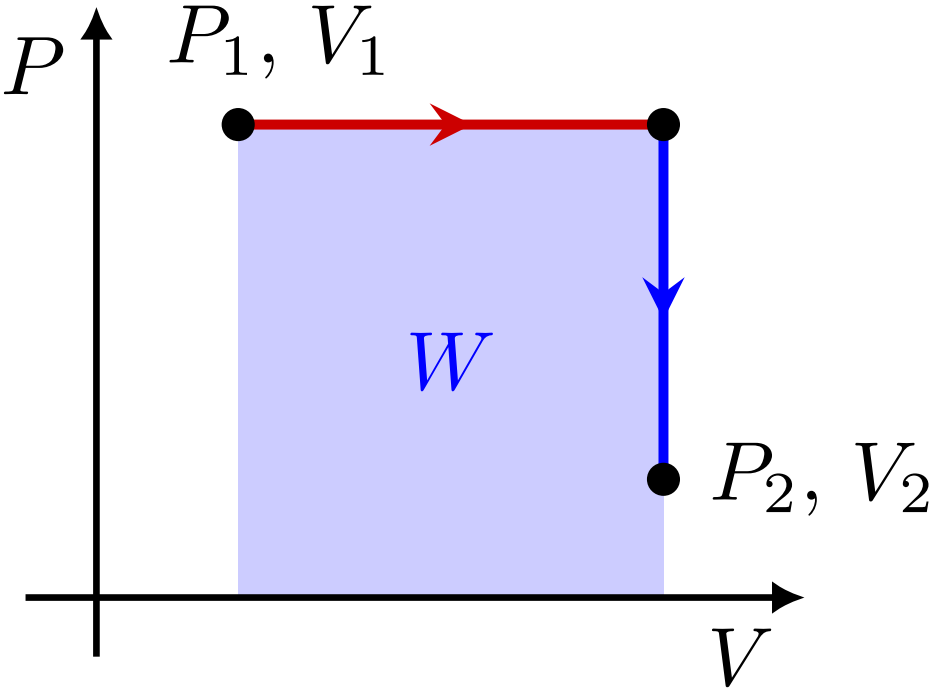
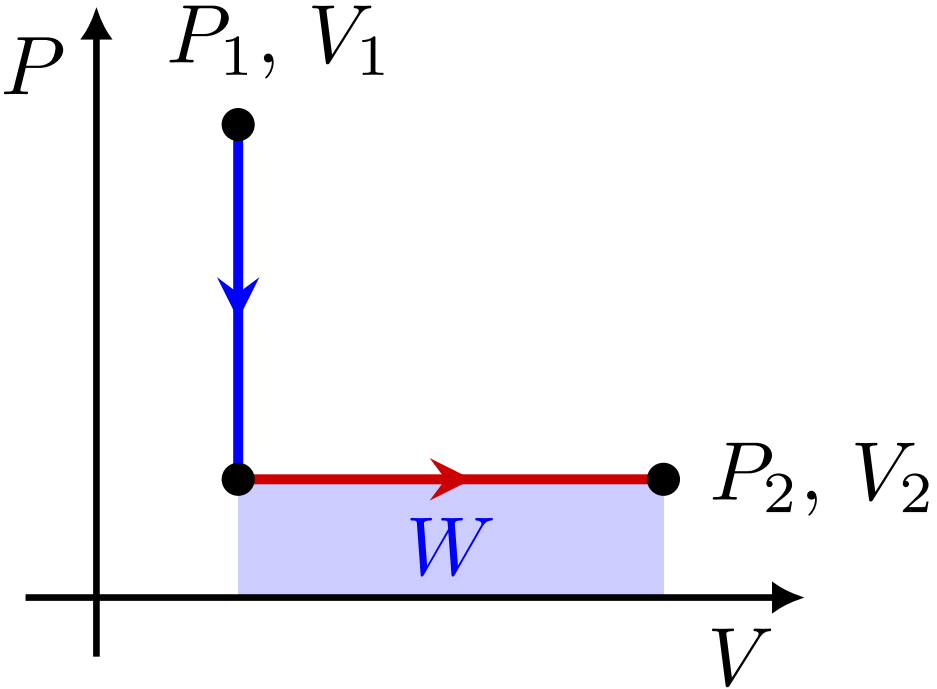
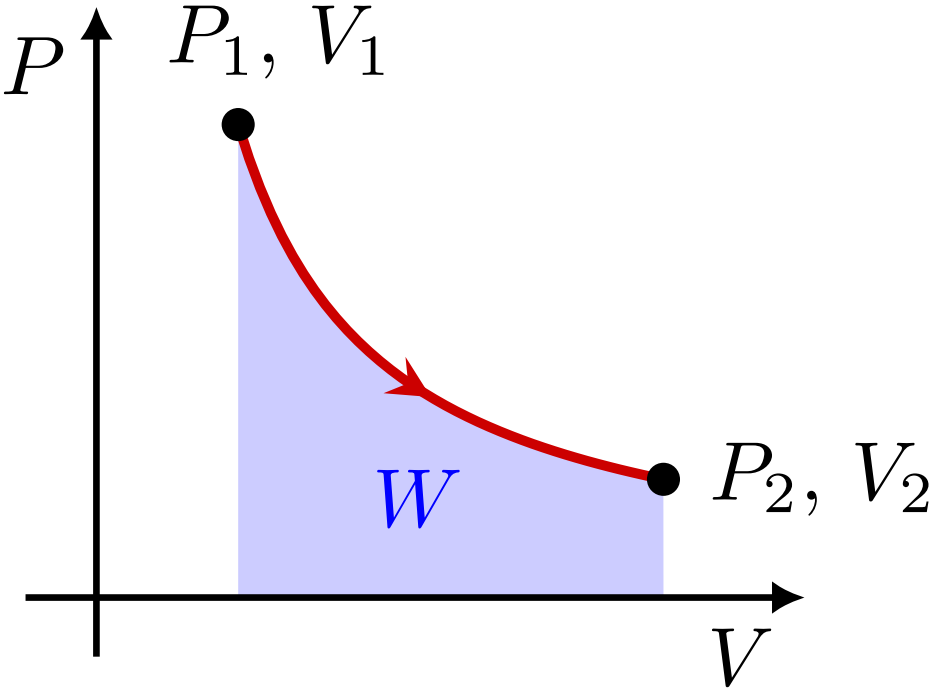
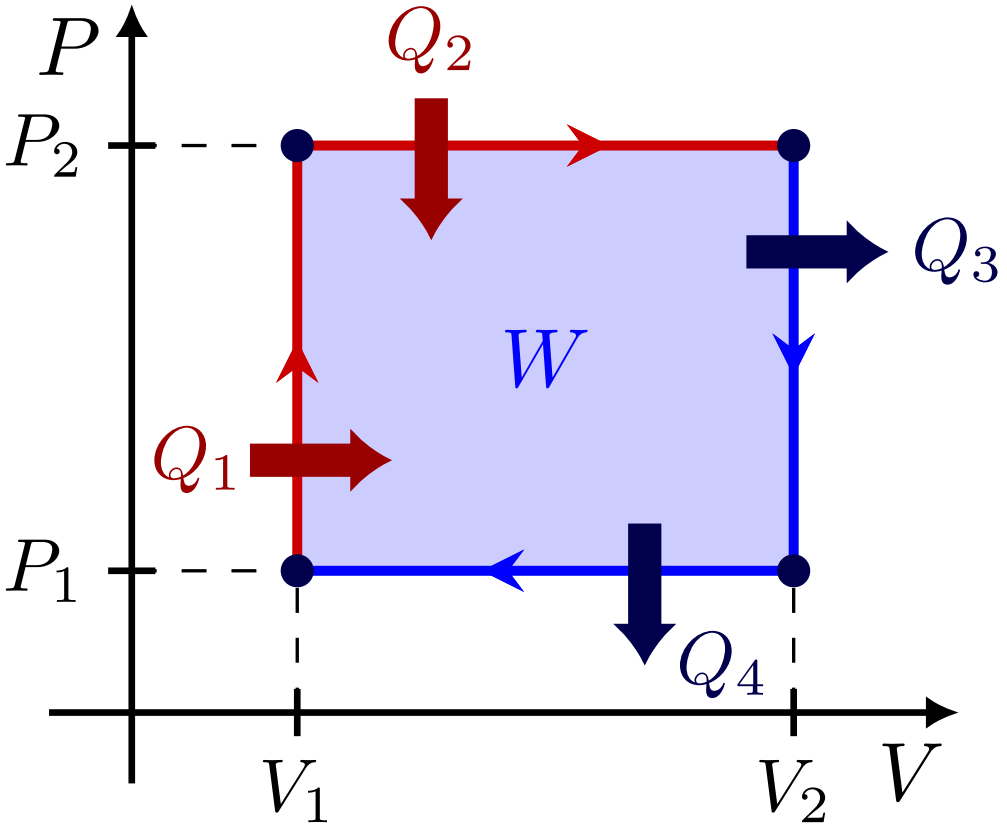
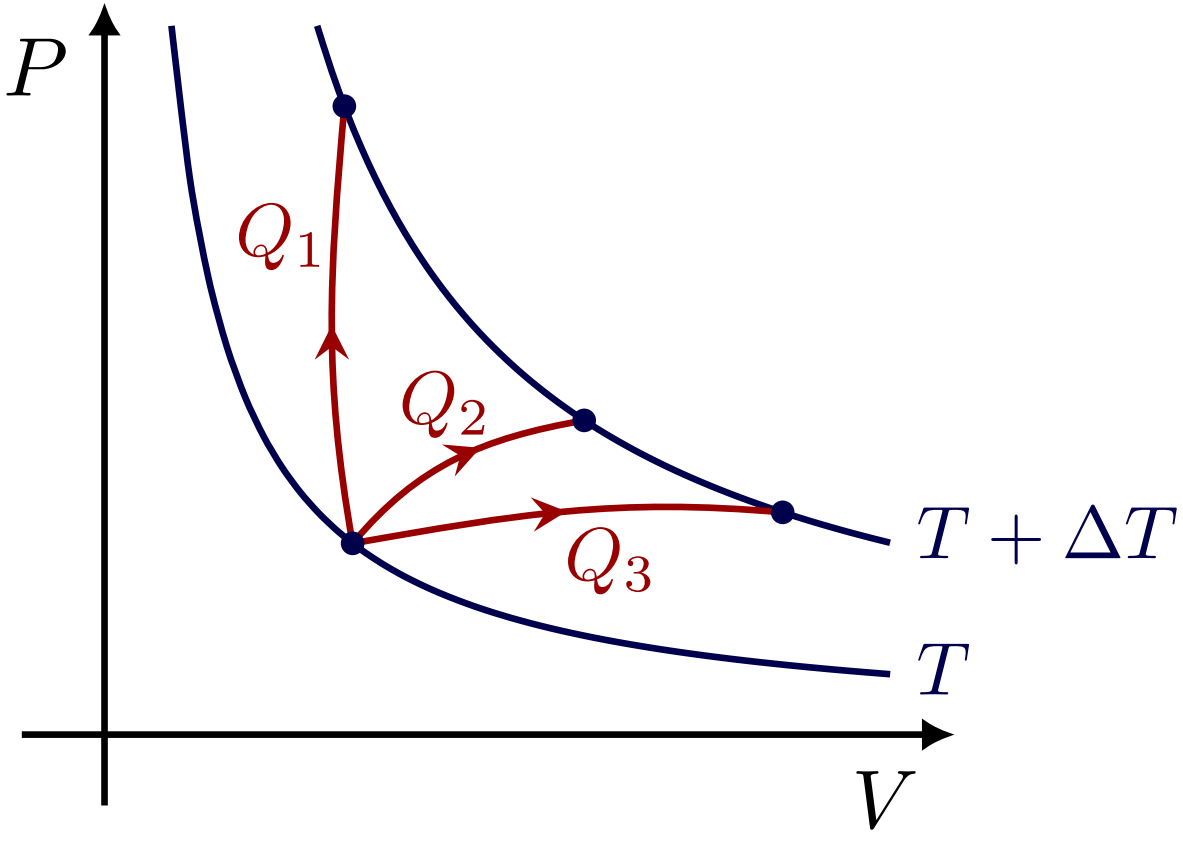
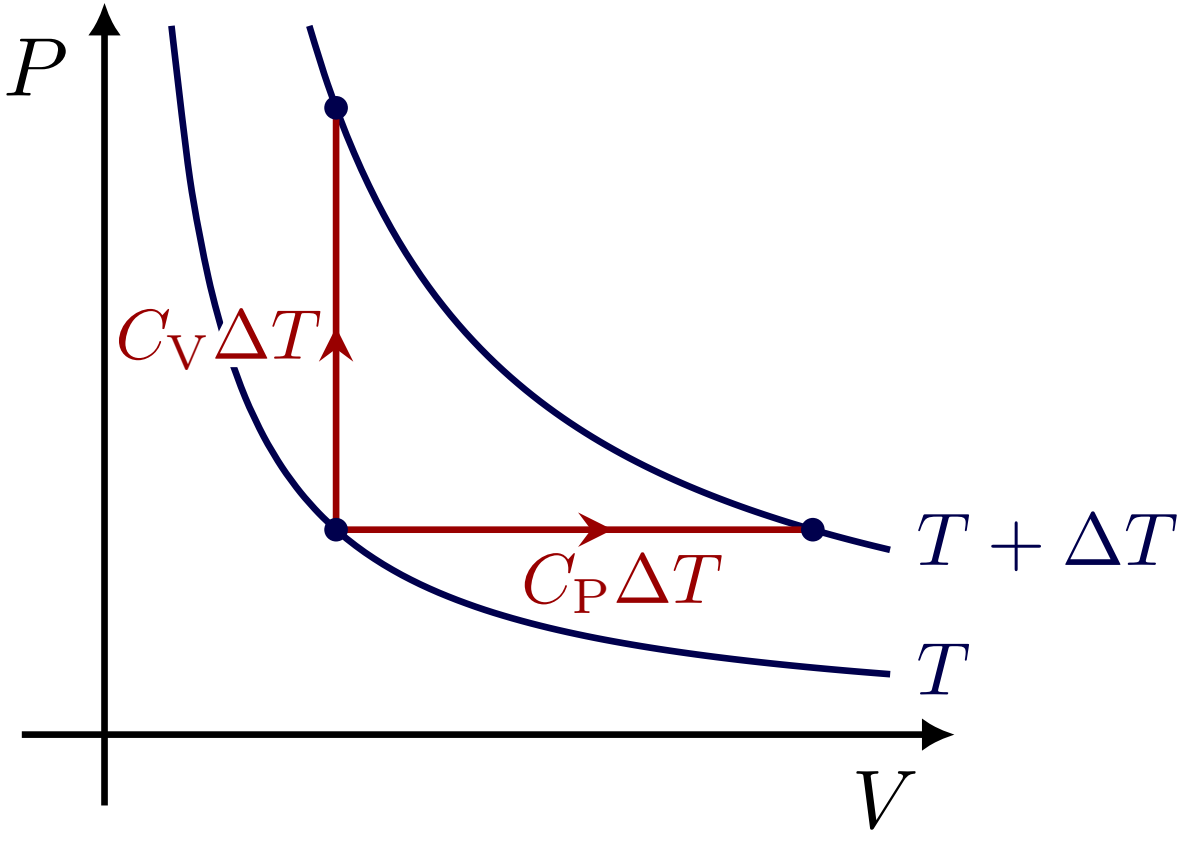

P
P_1, V_1
W
P_2, V_2
V

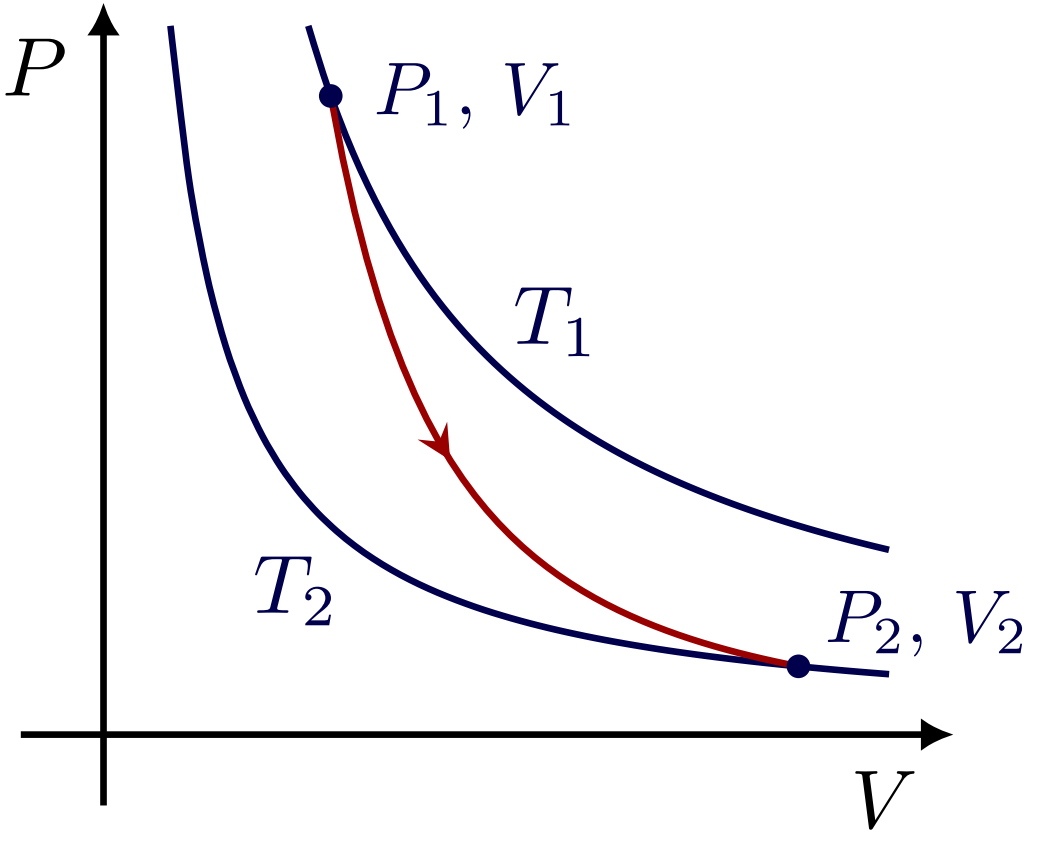

P
P_1, V_1
P_2, V_2
W
V

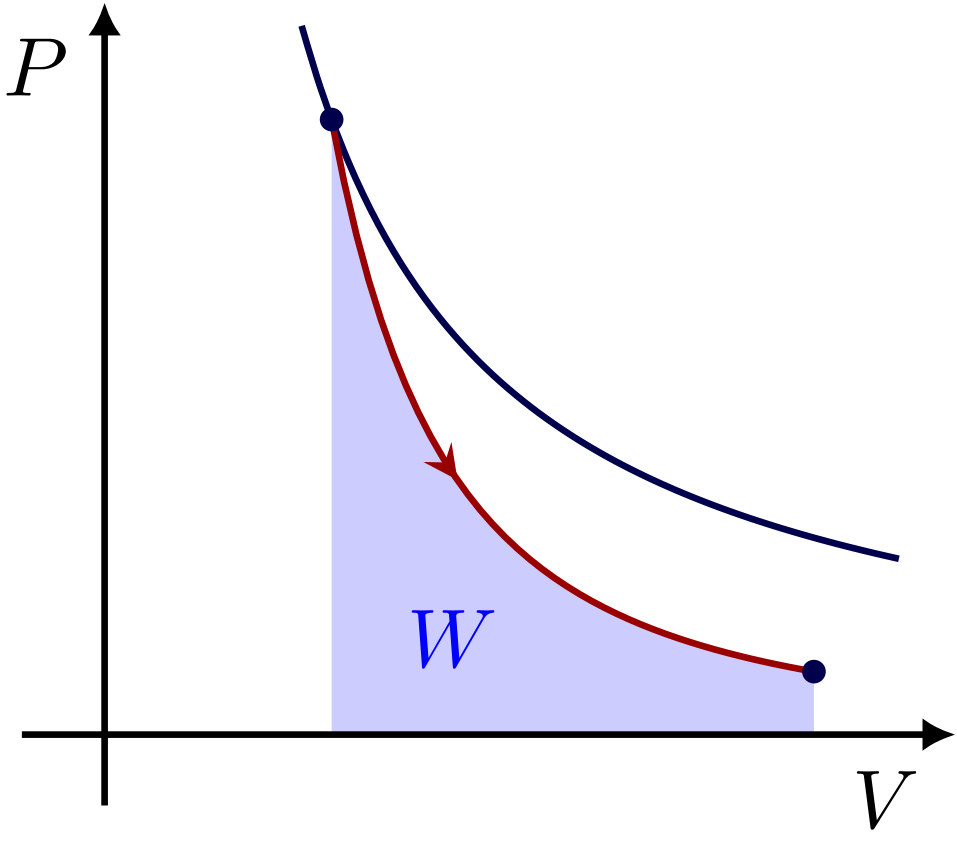

P
P_1, V_1
P_2, V_2
W
V

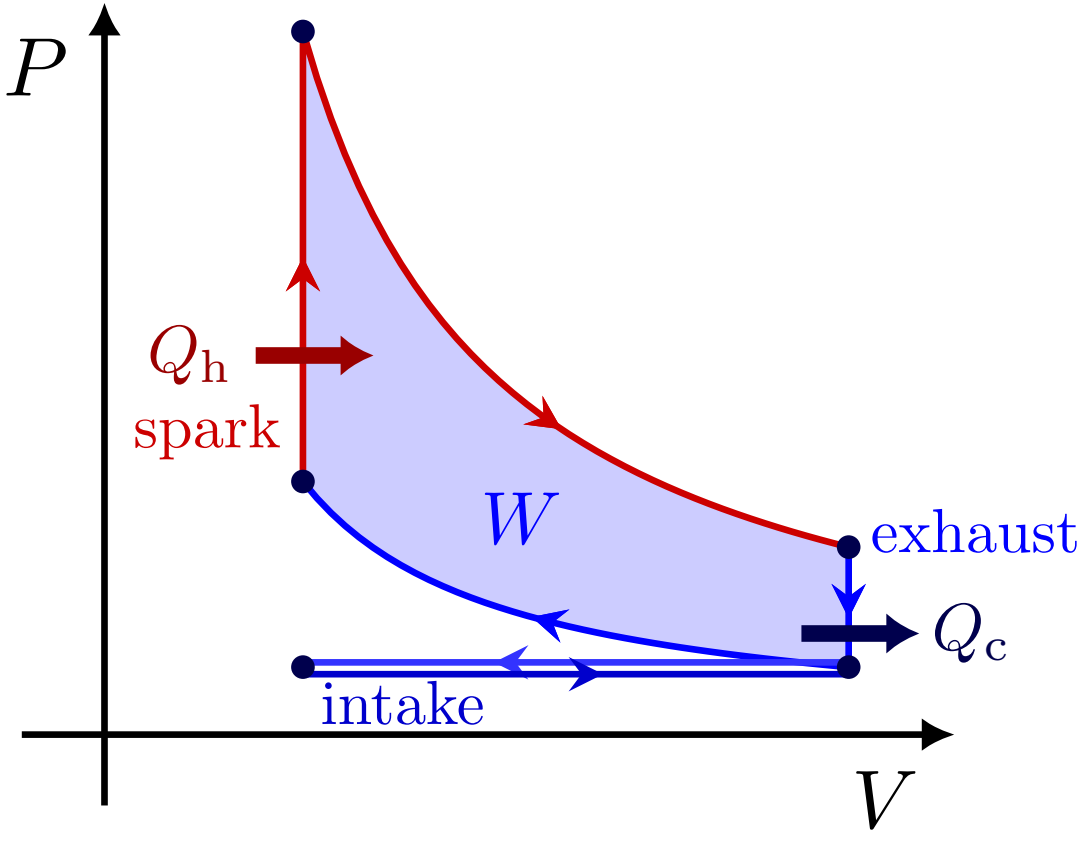

P
P_2
P_1
Q_2
Q_3
W
Q_1
Q_4
V_1
V_2
V

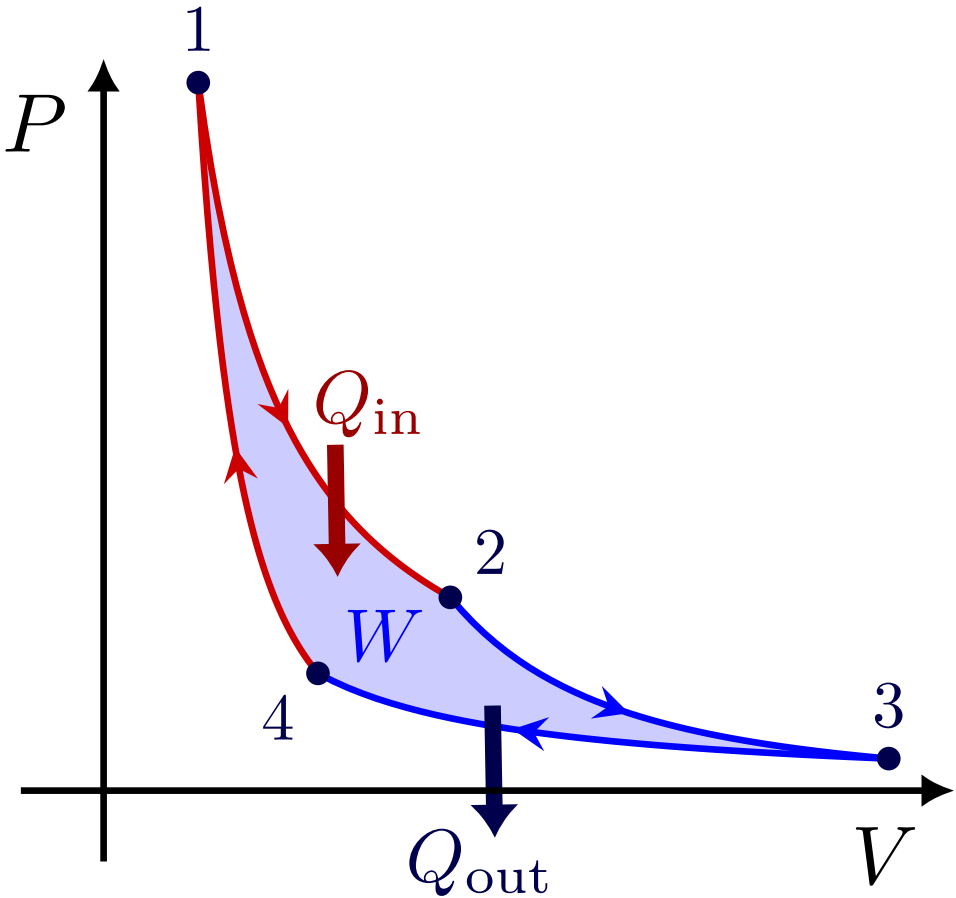

P
V
Q_1
Q_2
Q_3
$T + \Delta T$
T


P
$C_{\mathrm{V}}\Delta T$
$C_{\mathrm{P}}\Delta T$
$T + \Delta T$
T
V


P
V
P_1, V_1
P_2, V_2
T_1
T_2


P
W
V


P
Q_{h}
spark
W
exhaust
Q_{c}
intake
V


1
P
Q_{in}
2
W
4
3
Q_{out}
V